many of the checks against your services and the "outside view"—that is, customer experience (CX) as possible. In this context, it's crucial to emphasize this outside view; imagine a situation in which all your services report they are healthy and working as expected, but your end users can't access the app due to, say, a DNS issue.

The actual mechanisms you need to use to perform health and performance monitoring are really a spectrum, including the following:

- *Reachability monitoring*—Perform a simple call (ICMP, TCP, HTTP) to check for reachability or to ensure an endpoint returns a predefined status code.
- *Synthetic monitoring*—Execute scripts to simulate typical end user actions. These simulated end users are sometimes called *canaries* (a reference to the original use case, in which those little birds were used to detect gas leaks in mines). Several open source and commercial offerings allow you to perform synthetic monitoring—for example, Datadog (https://docs.datadoghq.com/synthetics/, Elastic (http://mng.bz/5wDO), New Relic (http://mng.bz/6Dle), and Amazon CloudWatch (http://mng.bz/o17Z). See figure 7.1 for an example.

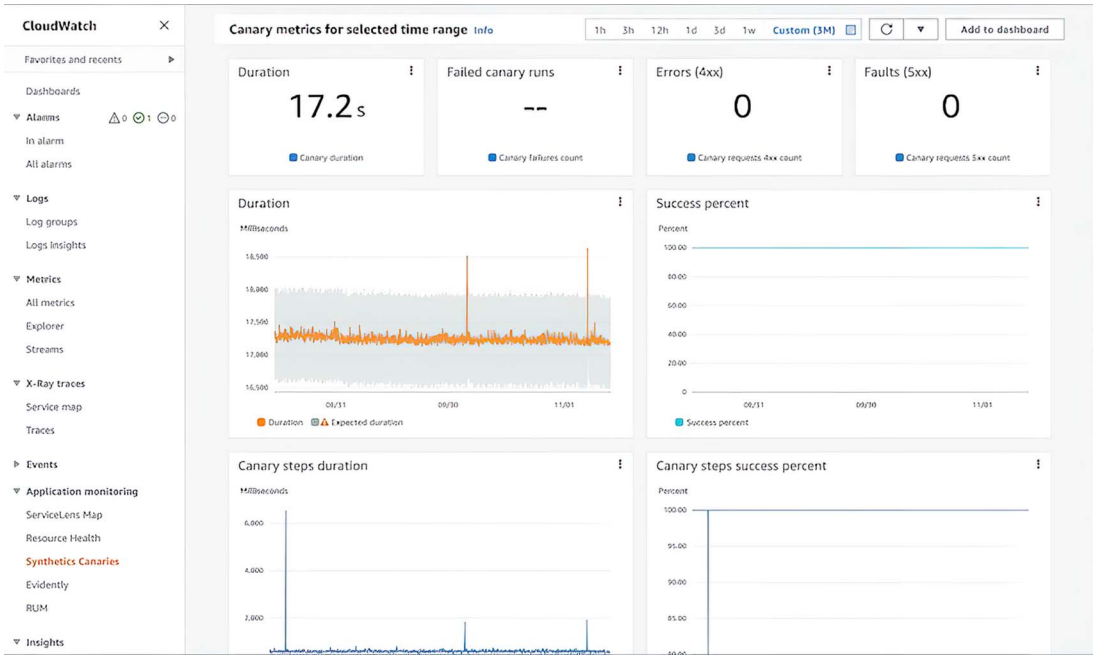


Figure 7.1 CloudWatch, an example of synthetic monitoring, tracking the availability of a website

You typically want to automate the process of checking for a condition and inform someone responsible about it, which is also known as *alerting* or *alarming*. There is a nice high-level article, "When to Alert on What?" (https://ali.sattari.me/posts/2023/when-to-alert-on-what/), which is a useful article for getting started on the topic,